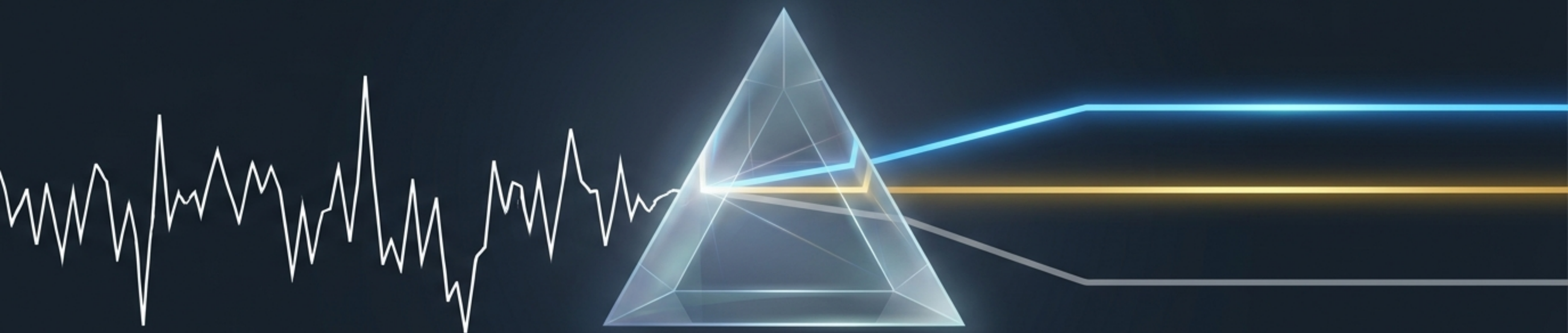
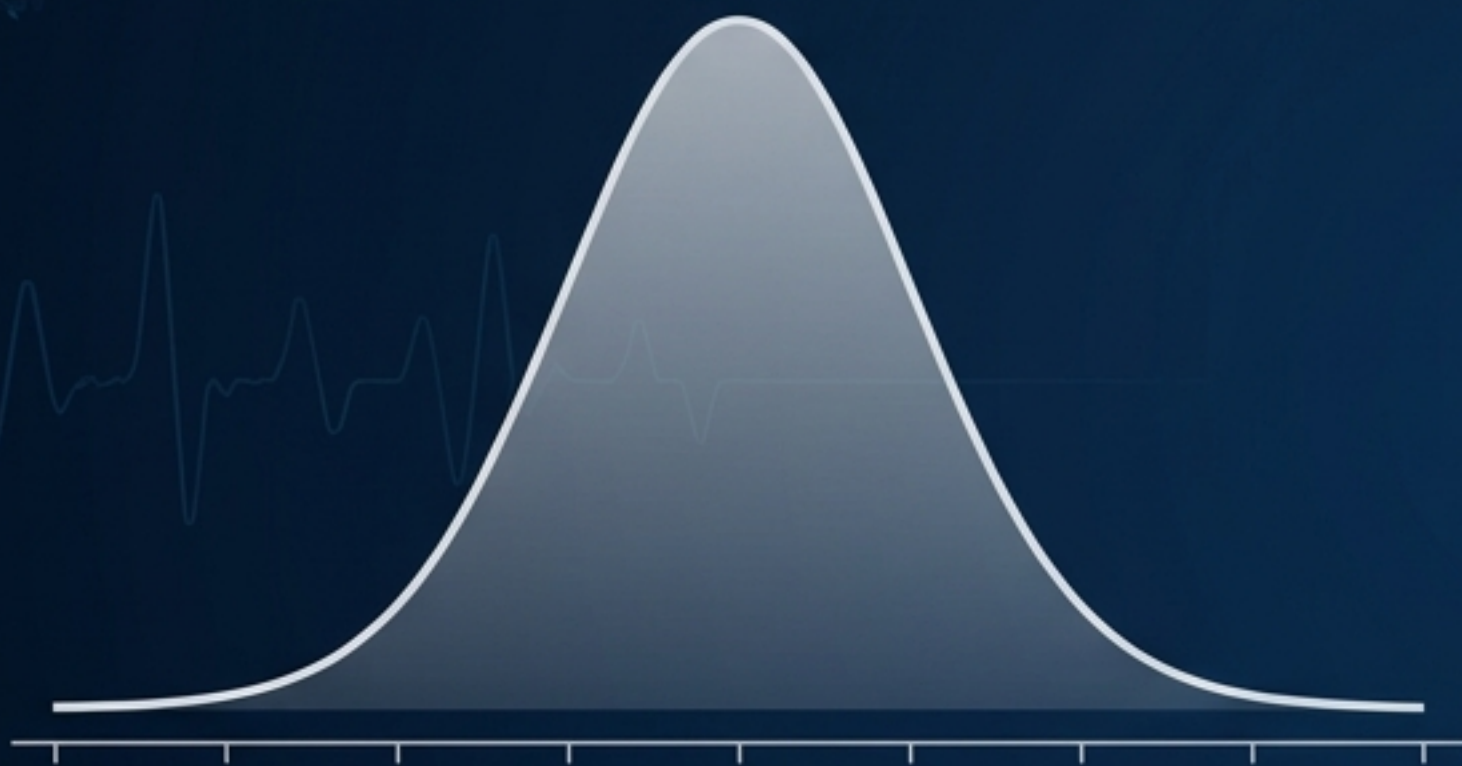


The Signal in the Noise: Architecting N-of-1 Trials for Precision Medicine



Population Means Obscure Individual Mechanistic Truths

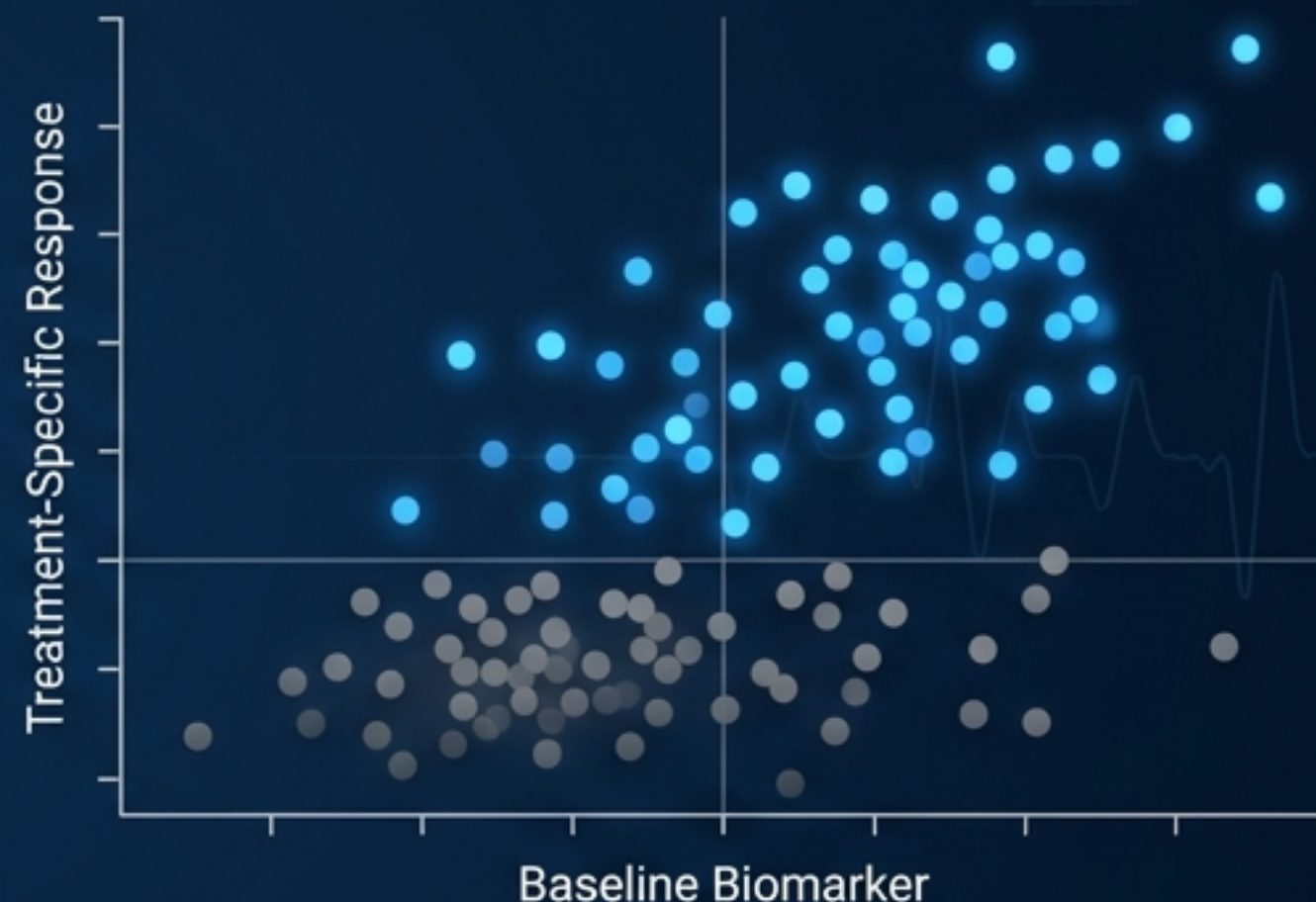
The Status Quo: Parallel-Group RCT



Evaluates "The Average Patient".

Flaw: Cannot distinguish if improvement is treatment-specific or identical to a placebo response. Obscures Heterogeneity of Treatment Effect (HTE).

The Future: Aggregated N-of-1



Evaluates "The Specific Patient".

Advantage: Quantifies the exact relationship between a baseline biomarker and treatment-specific response, rescuing effective drugs that fail population averages.

The N-of-1 Paradigm Isolates Individual Treatment Effects at Scale



Single Patient

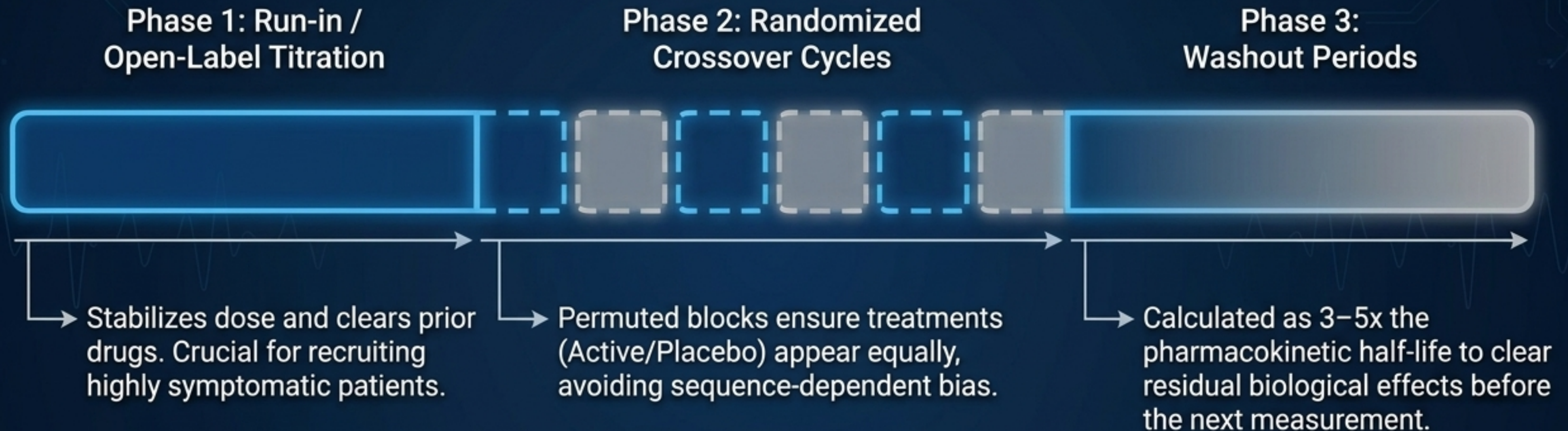
Acts as their own control across multiple crossover cycles, establishing an individualized baseline and true within-subject variance.



Aggregated N-of-1

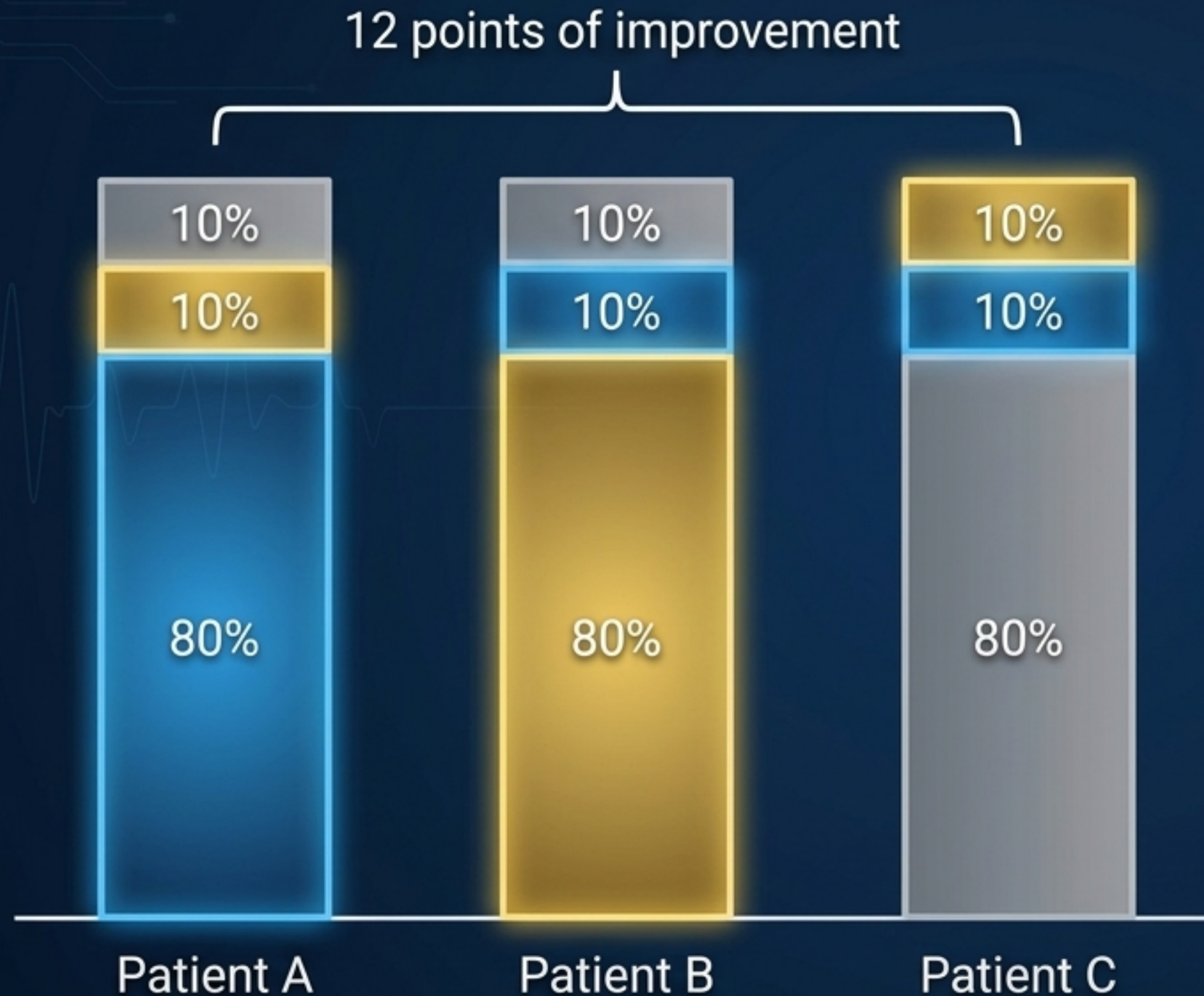
Pooling these individual trials using Linear Mixed-Effects Models (LME) generates robust population-level inferences while sharpening individual estimates through Empirical Bayes shrinkage.

Trial Mechanics Turn Temporal Sequences into Instruments of Measurement



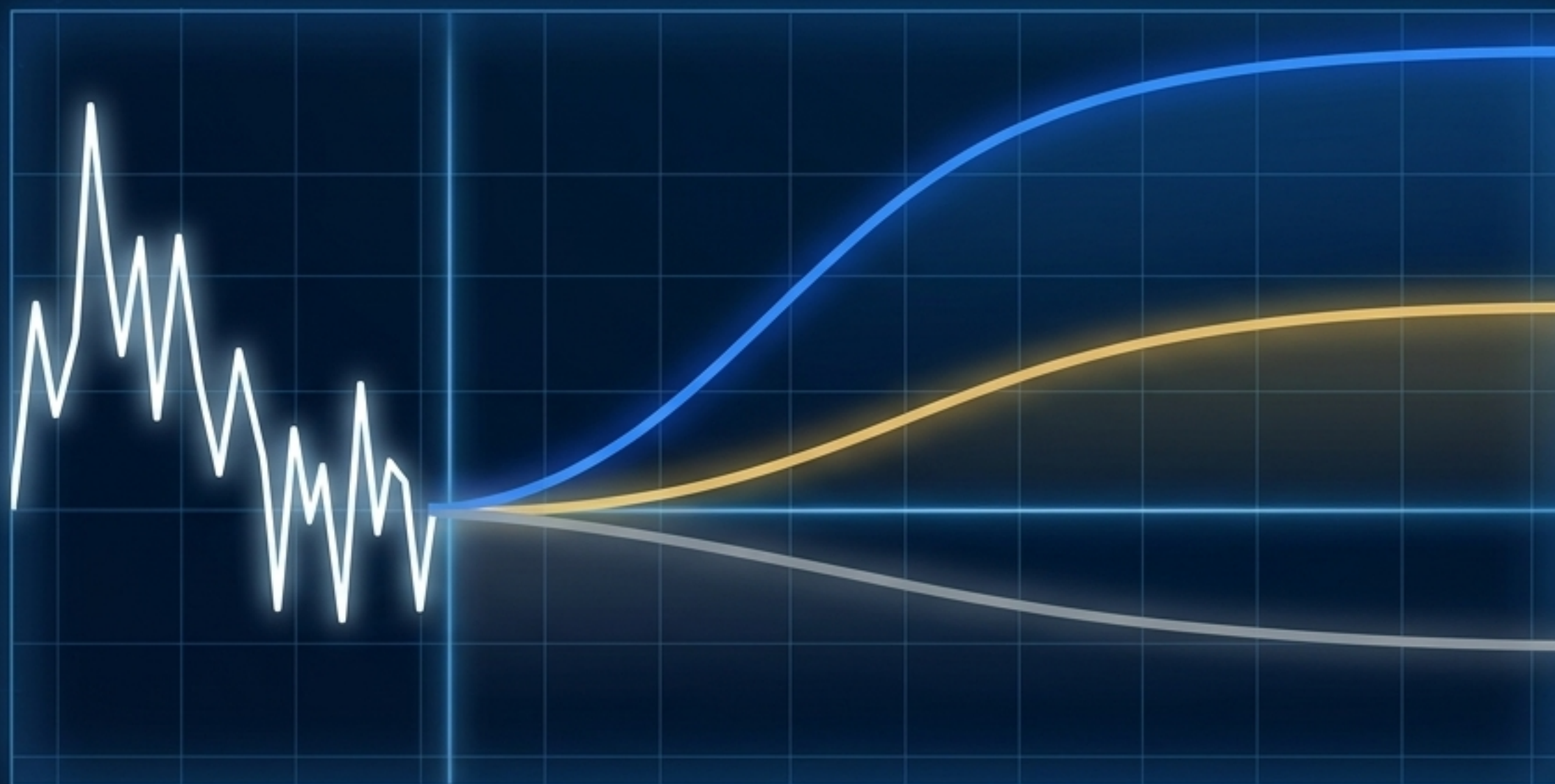
Key Insight: N-of-1 trials exist to exploit within-patient temporal structure. Serial correlation (modeled via corCAR1 continuous-time autocorrelation) must be actively managed, not flattened.

A Single Point of Symptom Improvement Hides Three Different Causes



- Traditional 1-component models ($Y = \text{Treatment} + \text{Error}$) create a wastebasket of bias. They falsely attribute psychological expectancy and natural history to pharmacological efficacy.
- Without separating these causes, validating a purely biological biomarker is mathematically impossible.

The BR-PB-TV Framework Mathematically Deconstructs the Symptom Waveform



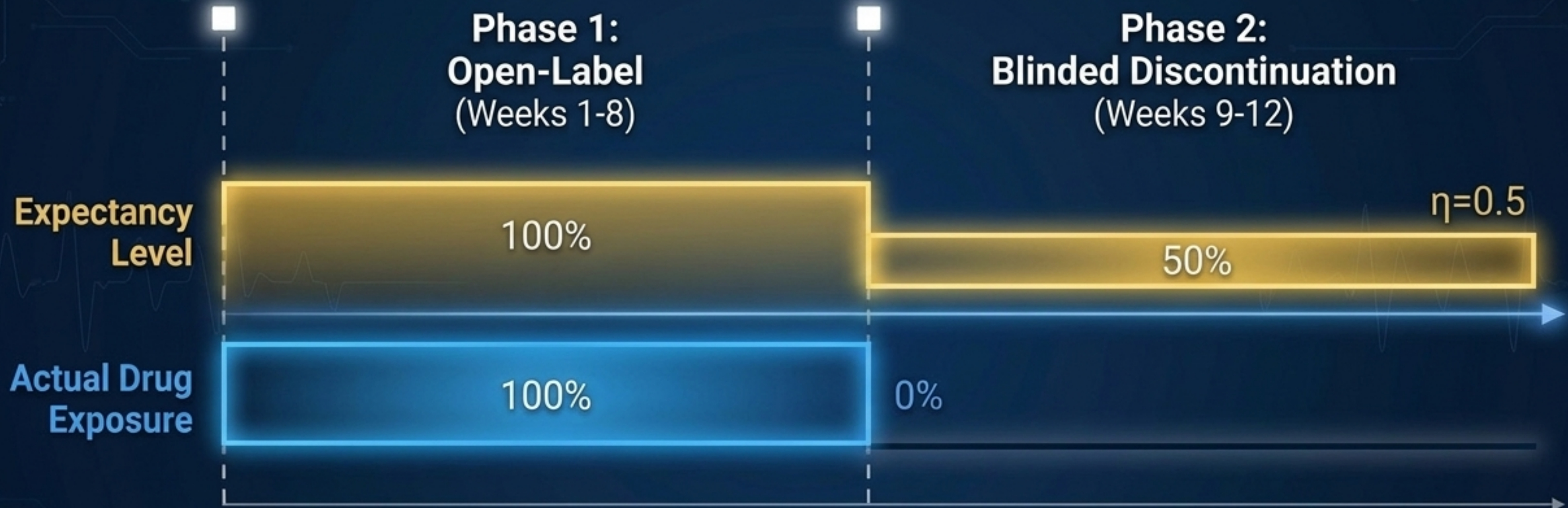
Biological Response (BR):
The true pharmacological effect.
Driven entirely by cumulative time
on active drug.

Placebo-Belief (PB):
The psychological effect.
Driven by time under positive
expectancy, scaled by an
expectancy multiplier (η).

Time-Varying (TV):
Natural history and regression to the
mean. Driven strictly by calendar
time since study entry.

$$Y_{it} = BL_i - [BR_{it} + PB_{it} + TV_{it}] + \epsilon_{it}$$

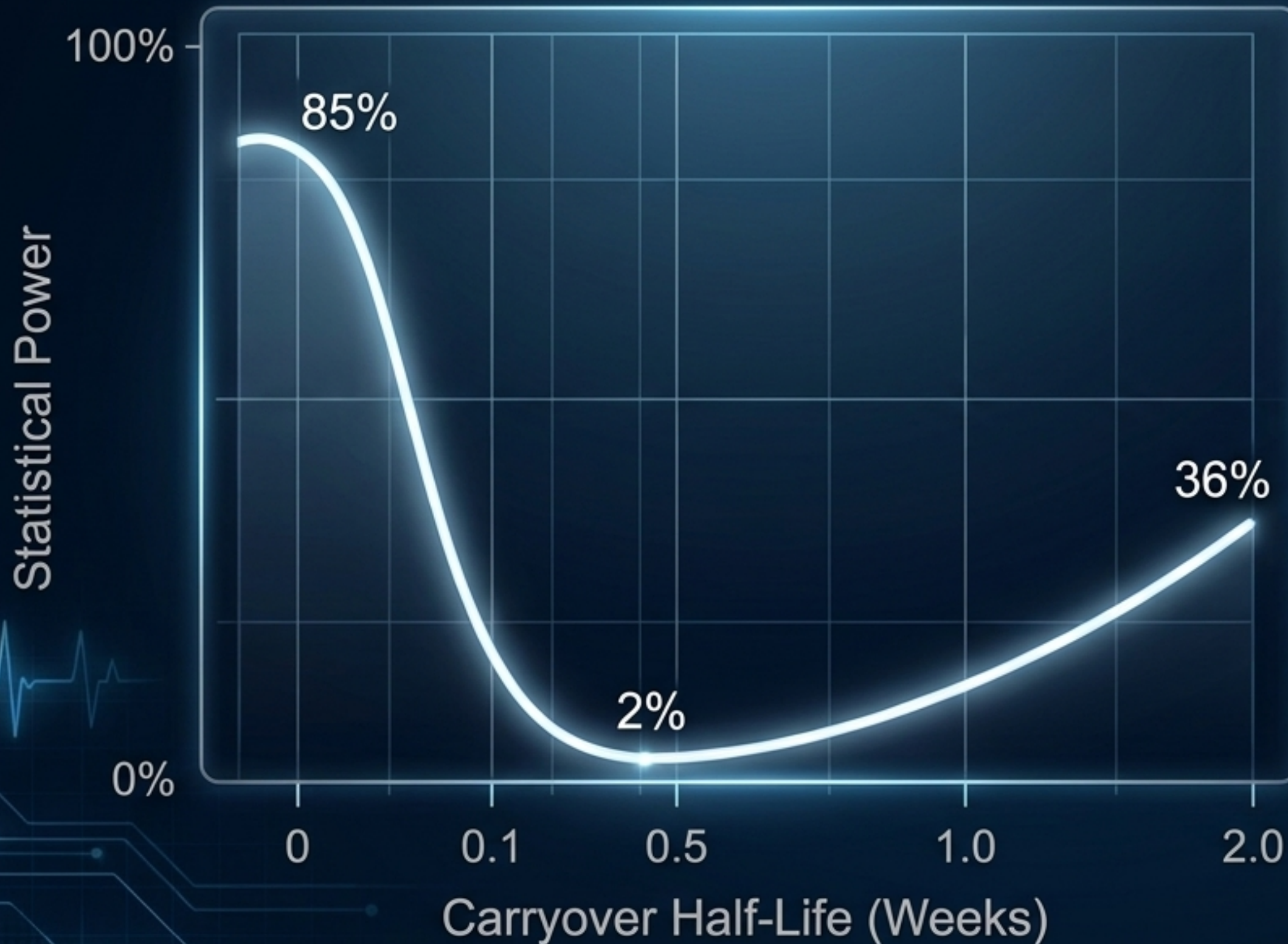
Hybrid Trial Designs Engineer Statistical Identifiability



Patient knows they receive the drug.
BR and PB rise together
(statistically indistinguishable).

Patient is told they may be on placebo.
Biological Response crashes, but
Placebo-Belief persists. This engineered
gap perfectly isolates the true BR.

Unmodeled Carryover Effects Create a Hidden Power Trap



The Paradox:

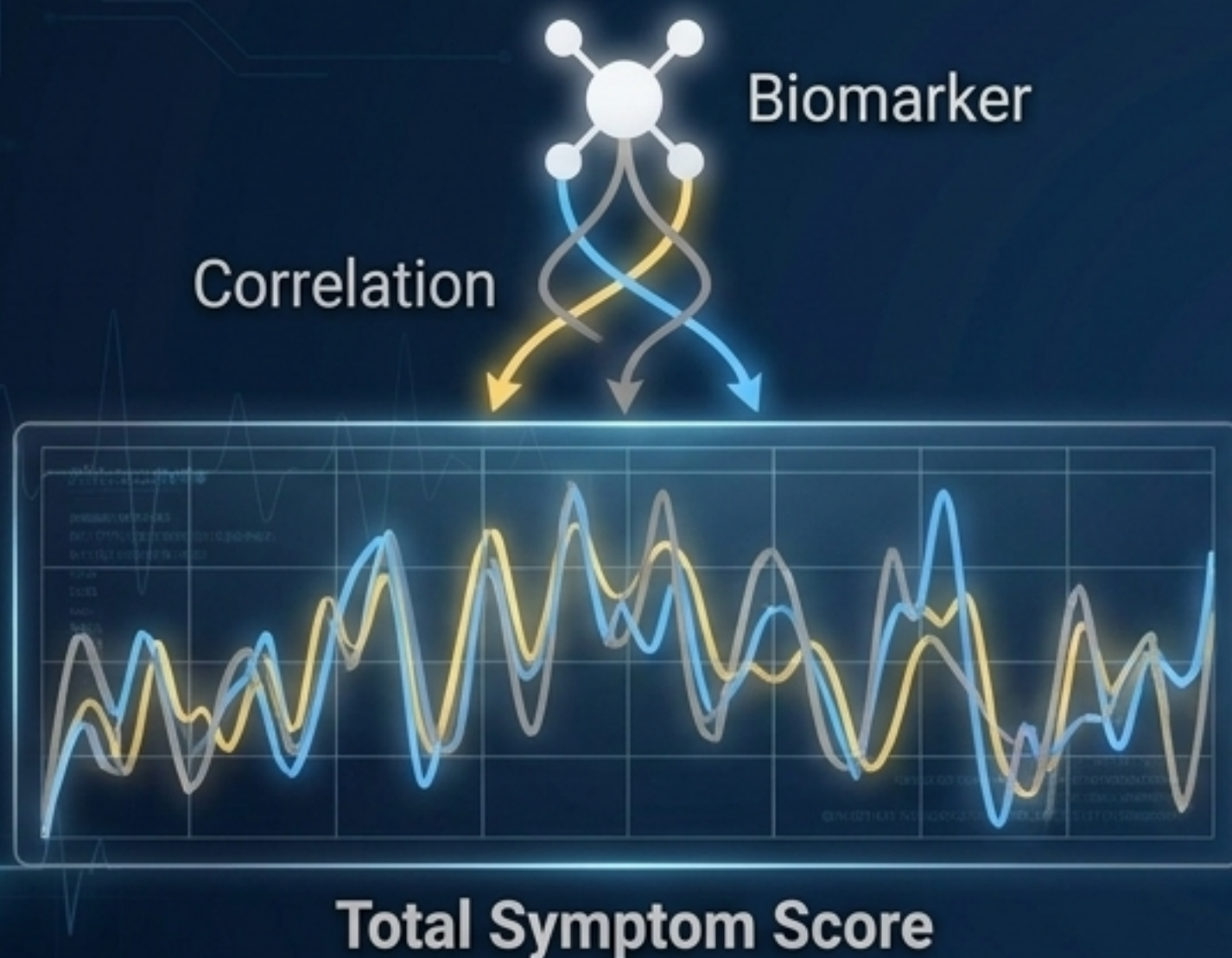
Very short carryover (0.1–0.5 weeks) is the worst-case scenario. The true effect decays rapidly, maximizing unexplained variance against a constant analysis predictor.

The Rule:

You must strictly engineer washouts (5x half-life) or explicitly model the exponential carryover decay.

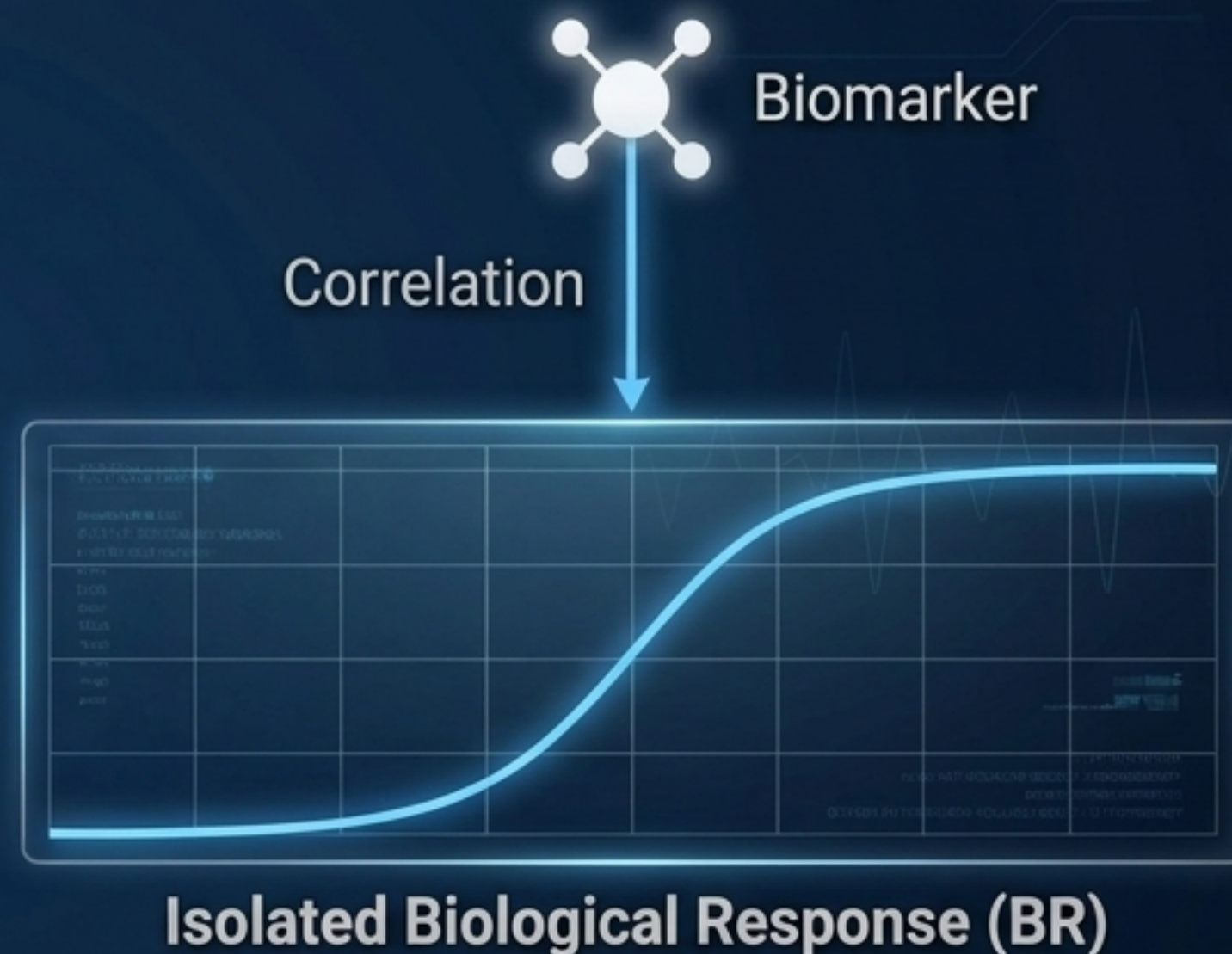
True Predictive Biomarkers Must Isolate the Biological Response

Panel A: The Contaminated Signal



If a biomarker predicts total response, it may merely predict a highly suggestible patient (Placebo) or natural disease drift (TV). It is clinically useless for drug targeting.

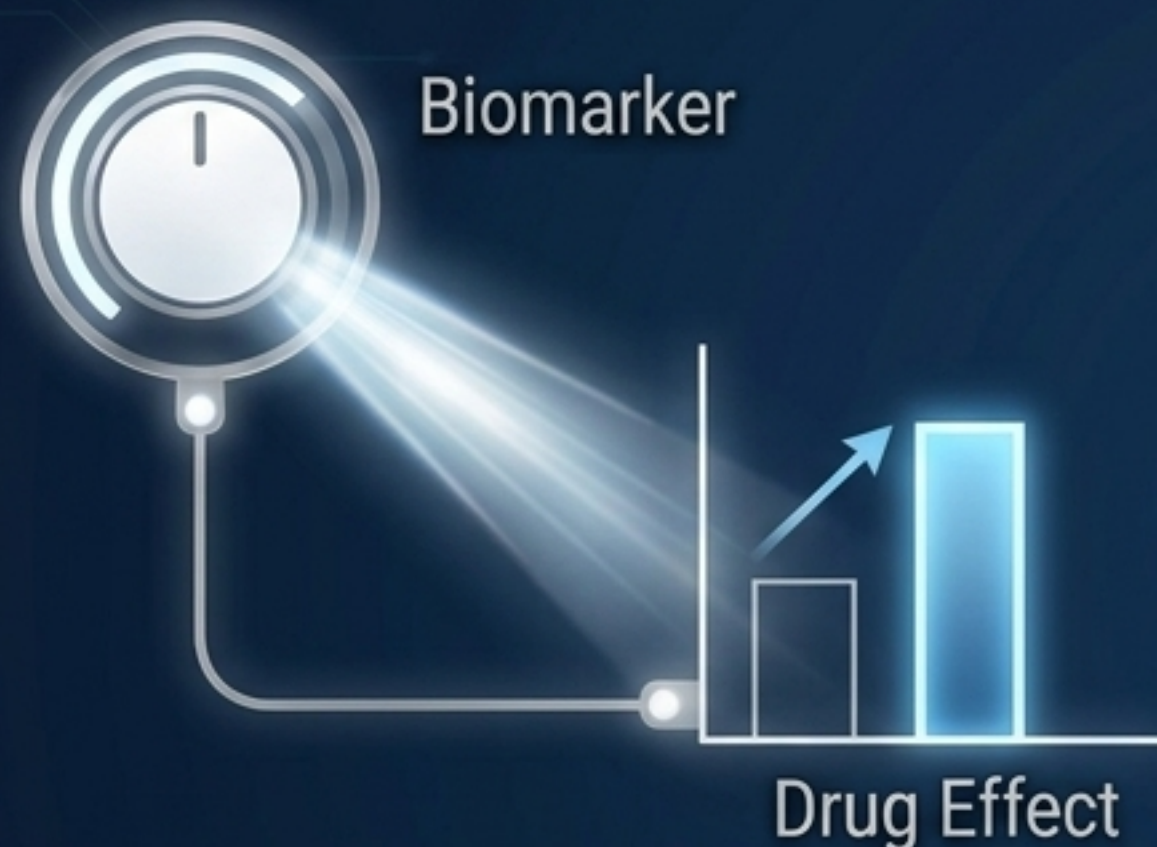
Panel B: The Pharmacological Truth



The BR-PB-TV decomposition allows us to test the biomarker strictly against the physiological drug mechanism, ensuring predictive validity.

Data-Generating Architectures Dictate Carryover Vulnerability

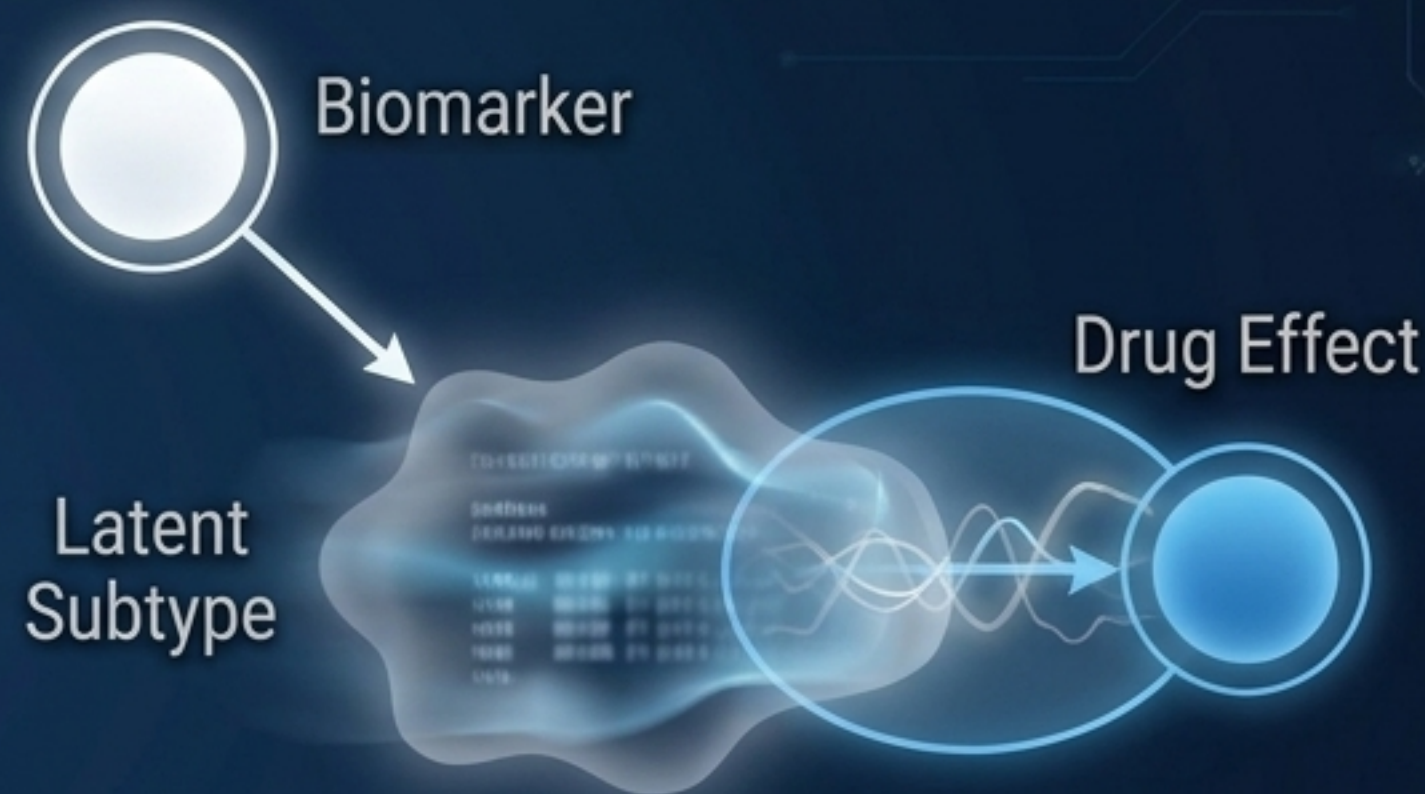
Architecture A: Direct Mean Moderation



Mechanism: The biomarker deterministically scales the magnitude of the effect (e.g., eGFR determining drug clearance).

Impact: Robust. Only 10-15% power loss under carryover.

Architecture B: Differential Correlation (MVN)



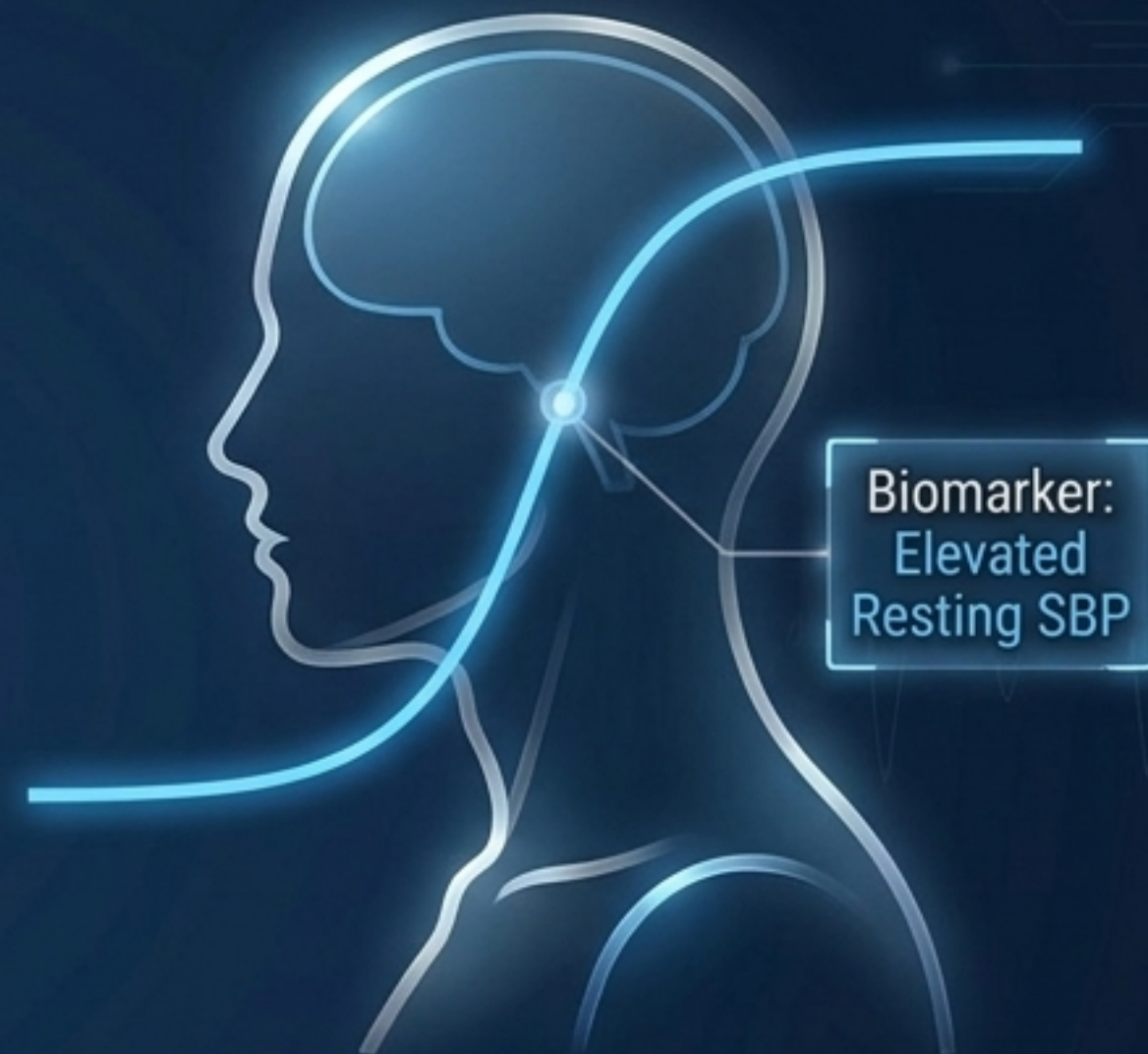
Mechanism: The biomarker predicts who will respond based on an underlying state (e.g., Blood Pressure indexing a noradrenergic PTSD subtype).

Impact: Vulnerable. 40-60% power loss under carryover because the differential correlation signal physically erodes during washout.

Advanced Modeling Rescues Effective Drugs from Failed Trials



Resolving carryover paradoxes and isolating biological truth requires deliberate statistical architecture.



The Prazosin for PTSD Case:

When parallel RCTs fail to show population means, N-of-1 identifies the precise neurobiological subgroup that benefits—delivering the definitive roadmap for true precision medicine.